Name - Shashank Navad
ASUID- 1229407428
ASURITE User ID - snavad

HW2

Task 1 -

Random Forest outperforms Decision Tree in terms of accuracy and robustness. This is because Random Forest combines multiple decision trees, reducing overfitting and improving overall performance. The Random Forest model typically achieves higher accuracy, sensitivity, and specificity compared to a single Decision Tree.

However, the choice between these algorithms depends on specific use cases:

Decision Tree advantages:

- Easier to interpret and visualize
- Faster to train and make predictions
- Better for capturing linear relationships

Random Forest advantages:

- Higher accuracy and stability
- Lower risk of overfitting
- Better handling of high-dimensional data

The main drawbacks of Random Forest are reduced interpretability and increased computational complexity. Decision Trees, while more prone to overfitting, offer clearer insights into the decision-making process.

GitHub Code - https://github.com/shashnavad/Data_Mining/blob/main/HW2/HW2-Changes-ShashankNavad-v1.ipynb